strong (spacetime is so strongly warped) that time itself (every-one's time) flows into the singularity.[1] Since the astronaut, like anyone else, must move inexorably forward in time, he is driven with the flow of time into the singularity. No matter what he does, no matter how he blasts his rocket engines, the astronaut cannot avoid the singularity's infinite stretch and squeeze.

Whenever we physicists see our equations predict something infinite, we become suspicious of the equations. Almost nothing in the real Universe ever gets truly infinite (we think). Therefore, an infinity is almost always a sign of a mistake.

The singularity's infinite stretch and squeeze was no exception. Those few physicists who studied Oppenheimer and Snyder's publication during the 1950s and early 1960s agreed unanimously that something was wrong. But there the unanimity stopped.

One group, led vigorously by John Wheeler, identified the infinite stretch and squeeze as an unequivocal message that general relativity fails inside a black hole, at the endpoint of stellar implosion. Quantum mechanics should prevent tidal gravity from becoming truly infinite there, Wheeler asserted; but how? To learn the answer, Wheeler argued, would require marrying the laws of quantum mechanics with the laws of tidal gravity, that is, with Einstein's general relativistic laws of curved spacetime. The progeny of that marriage, the laws of quantum gravity, must govern the singularity, Wheeler claimed; and these new laws might create new physical phenomena inside the black hole, phenomena unlike any we have ever met.

A second group, led by Isaac Markovich Khalatnikov and Evgeny Michailovich Lifshitz (members of Lev Landau's Moscow research group), saw the infinite stretch and squeeze as a warning that Oppenheimer and Snyder's idealized model of an imploding star could not be trusted. Recall that Oppenheimer and Snyder required, as a foundation for their calculations, that the star be precisely spherical and nonspinning and have uniform density, zero pressure, no shock waves, no ejected matter, and no outpouring radiation (Figure 13.2). These extreme idealizations were responsible for the singularity, Khalatnikov and Lifshitz argued. Every real star has tiny, random deformations (tiny, random nonuniformities in its shape, velocity, density, and pressure), and as the star implodes, they